MISOPROSTOL

Teratogenic

Heat-stable — PPH in low-resource settings

CHEMICAL NAME Methyl (13E)-11α,16-dihydroxy-16-methyl-9-oxoprost-13-en-1-oate (synthetic prostaglandin E1 analogue)

CLASS Prostaglandin E1 analogue (uterotonic, gastroprotective)

TRADE NAMES Cytotec · Misodel (vaginal insert) · Angusta · Cyprostol

THERAPEUTIC USES / INDICATIONS

- Prevention and treatment of postpartum haemorrhage (when oxytocin unavailable).
- Induction of labour (cervical ripening); specialist protocols.
- Medical management of miscarriage and termination of pregnancy (with mifepristone) where legal.
- Prevention of NSAID-induced gastric ulcers.

ADMINISTRATION

- **PO**, **sublingual**, **buccal**, **vaginal** or **rectal** depending on indication.
- Induction: vaginal or oral low doses with continuous monitoring.
- Store dry; heat-stable.

DOSAGES

- **PPH prevention:** 600 mcg PO/sublingual after birth.
- **PPH treatment:** 800 mcg sublingual.
- **Labour induction:** 25 mcg vaginally every 4–6 h, or 25–50 mcg orally every 2–4 h (protocol-dependent).
- **Miscarriage (incomplete):** 400–600 mcg sublingual/PO single dose.
- **NSAID ulcer prophylaxis:** 200 mcg 2–4 times daily with food.

PHARMACOKINETICS

- **Onset:** sublingual 10–15 min; oral 30 min; vaginal 20 min (longer duration).
- Rapidly de-esterified to active misoprostol acid.
- Excreted in urine. **t½:** 20–40 min.

PHARMACODYNAMICS

- Binds **prostaglandin EP2/EP3 receptors** → uterine contractions, cervical softening and dilatation.
- In stomach: ↓ acid secretion, ↑ mucus and bicarbonate (cytoprotection).

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Diarrhoea, abdominal cramps, nausea, vomiting.
- Shivering and fever (dose-related, common with PPH doses).
- Headache.

ADVERSE EFFECTS (SERIOUS)

- Uterine hyperstimulation, uterine rupture (especially with previous caesarean scar).
- Teratogenicity (Möbius syndrome, limb defects) if pregnancy continues.
- Severe haemorrhage (incomplete abortion).

CONTRAINDICATIONS & CAUTIONS

- Pregnancy where continuation is intended (for ulcer use); women of childbearing age need contraception.
- Previous caesarean section/uterine scar (for induction at term) — relative.
- Hypersensitivity to prostaglandins.

KEY INTERACTIONS

- **Oxytocin:** wait ≥4 h after last misoprostol dose before starting oxytocin.
- **Magnesium-containing antacids:** ↑ diarrhoea.

NURSING MANAGEMENT

- Labour: continuous fetal and contraction monitoring.
- PPH: assess uterine tone, blood loss, vital signs; monitor temperature (shivering/fever expected).
- Check pregnancy status before GI use.
- Monitor for diarrhoea, dehydration.

PATIENT EDUCATION

- Shivering and fever after the dose usually settle within hours.
- Report heavy bleeding, severe pain or fever lasting >24 h.
- Do not use for stomach protection if pregnant or planning pregnancy.

KEY POINTS

- PGE1 analogue — uterotonic and gastroprotective.
- Shivering, fever, diarrhoea.
- PPH: 600 mcg prevention, 800 mcg treatment.
- Teratogenic; uterine rupture risk with scars.